# Dry Bulk Market Diagnostic Framework

*A simple data-driven view of Baltic Dry market conditions using BDI, iron ore and Brent crude*

## 1. Executive Summary

This note builds a simple way to read the dry bulk freight market using three public market series: the Baltic Dry Index, iron ore futures and Brent crude oil.

The starting idea is straightforward. Iron ore is a major dry bulk cargo. If iron ore demand is strong, it should support cargo flows, raise vessel demand and help freight rates. That is the economic story I wanted to test.

The result is more cautious. In the recent sample used here, iron ore momentum does not explain short-term BDI returns very well. The scatter plot does not show a clear relationship, and the correlation between daily BDI returns and 20-day iron ore momentum is close to zero.

My interpretation is not that iron ore is irrelevant. It is that the dry bulk market is not a one-driver market. Freight rates are affected by demand, but also by vessel supply, port congestion, seasonality, route-specific dynamics, fuel costs and broader market conditions.

The current market regime is assessed as Neutral. BDI is still above its 60-day moving average, so the medium-term trend remains supportive. At the same time, BDI has fallen below its 20-day moving average, iron ore momentum is negative, and the index is around 7.6% below its recent high. This points to a market that is not in stress, but has lost some short-term strength.

This framework is not meant to be a trading model. It is a market diagnostic. It helps separate trend, risk, demand-side support and cost-side pressure, so the market discussion becomes more structured.

## 2. Market Background

The Baltic Dry Index is used as the main market indicator. It gives a broad view of dry bulk freight conditions across key vessel segments.

Iron ore futures are used as a demand-side indicator. The logic is that iron ore is one of the most important dry bulk cargoes. If iron ore demand strengthens, this should in principle support freight demand.

Brent crude oil is used as a simple cost-side indicator. It is not the same as bunker fuel, but it gives a practical view of broader energy cost pressure.

The first-principles logic is:

- Iron ore demand -> cargo flows -> vessel demand -> freight rates -> BDI
- Brent crude -> energy cost pressure -> operating cost / market pressure -> freight conditions

This is intentionally a simple framework. The goal is not to capture every driver of the market. The goal is to test whether a few interpretable indicators can give a useful first view of dry bulk freight conditions.

## 3. Research Question and Hypothesis

The main research question is: What drives dry bulk freight market conditions?

Version 1 tests one simple hypothesis: iron ore demand helps explain freight market movements.

This hypothesis makes economic sense, but it still needs to be tested. The purpose is not to prove a perfect causal relationship. The purpose is to see whether iron ore momentum alone gives useful information about short-term BDI movements.

## 4. Data and Methodology

The analysis uses daily data for BDI, iron ore futures and Brent crude oil. The series are cleaned, merged by date and forward-filled where market calendars differ. Returns and rolling indicators are then calculated.

The framework uses the following indicators:

- BDI daily returns
- BDI 20-day and 60-day moving averages
- BDI 20-day annualized volatility
- BDI drawdown from recent peak
- Iron ore 20-day momentum
- Brent 20-day momentum

The market regime is classified using a simple rule-based view:

- Supportive: BDI is above both moving averages, iron ore momentum is positive, and cost pressure is not extreme.
- Caution: BDI is below both its 20-day and 60-day moving averages.
- Risk Watch: volatility is high or drawdown passes the stress threshold.
- Neutral: the signals are mixed.

This rule-based approach is deliberately transparent. It is easier to discuss and challenge than a black-box score, especially when the goal is market diagnosis rather than automated trading.

The data window is an important limitation. The analysis mainly uses around one recent year of data. This means the results should be read as a short-term diagnostic view of the current market period, not as a long-run structural study of dry bulk cycles. The sample does not cover major historical stress periods such as the first phase of the COVID-19 supply-chain shock or the initial Russia-Ukraine energy shock.

## 5. Current Market Diagnosis

The latest observation gives the following market state:

*Table 1: Current market diagnosis.*

| Metric | Value |
|---|---|
| BDI | 2,981 |
| 20-day moving average | 3,095.2 |
| 60-day moving average | 2,624.53 |
| 20-day annualized volatility | 31.19% |
| Drawdown from recent high | -7.59% |
| Iron ore 20-day momentum | -9.31% |
| Brent 20-day momentum | -7.37% |
| Market regime | Neutral |

The market picture is mixed. BDI is still above the 60-day moving average, which supports a positive medium-term view. But BDI is now below the 20-day moving average, so the short-term picture has weakened.

The drawdown is around -7.6%. That is meaningful, but it is much smaller than the deeper drawdown seen earlier in the sample. Therefore, the current market does not look like a stress regime.

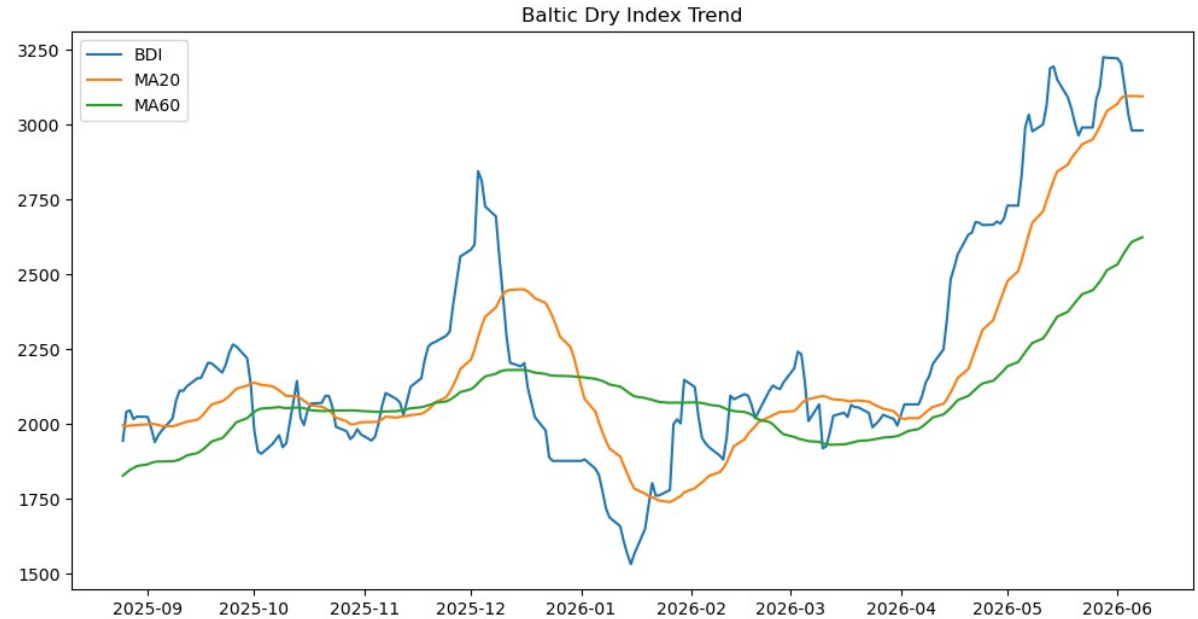
Iron ore momentum is negative, which points to weaker demand-side support. Brent momentum is also negative. That may reduce cost-side pressure, but it does not directly cancel the weaker demand signal.

In simple words: the market is still supported in the medium term, but it has lost some short-term strength.

## 6. Evidence

### 6.1 BDI trend

*Figure 1: BDI trend with 20-day and 60-day moving averages.*

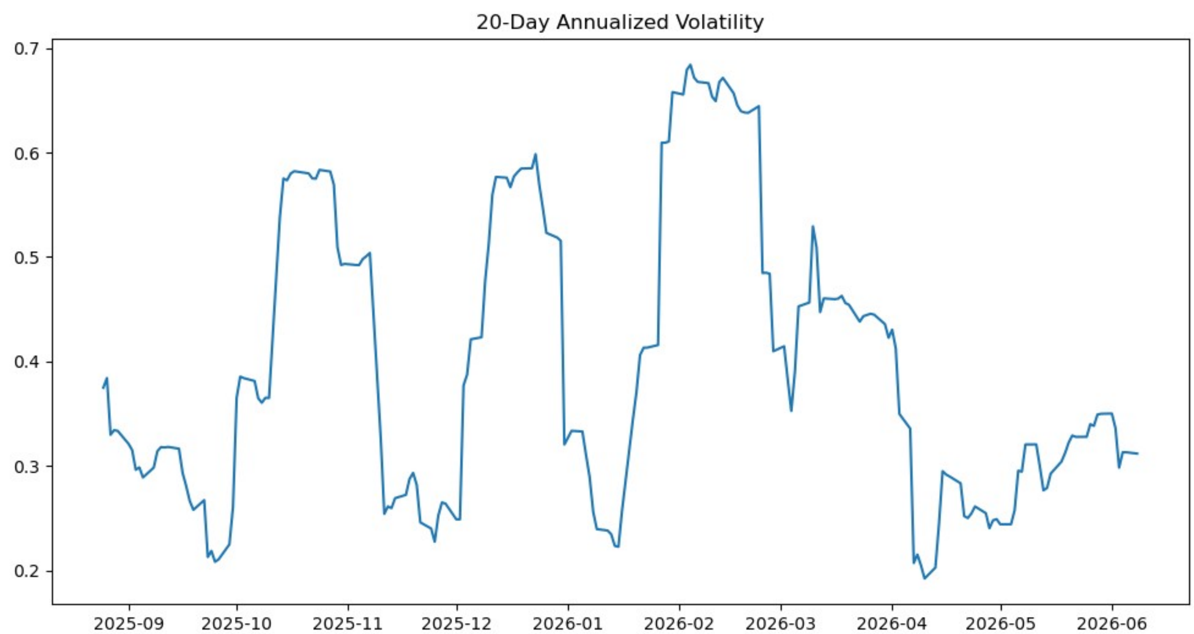



BDI recovered strongly from the early 2026 low and moved above 3,200 before pulling back. The chart gives two signals at the same time: BDI remains above the 60-day moving average, but it has moved below the 20-day moving average.

This is why I do not classify the market as clearly supportive or clearly cautious. The signals are mixed.

### 6.2 Volatility

*Figure 2: BDI 20-day annualized volatility.*

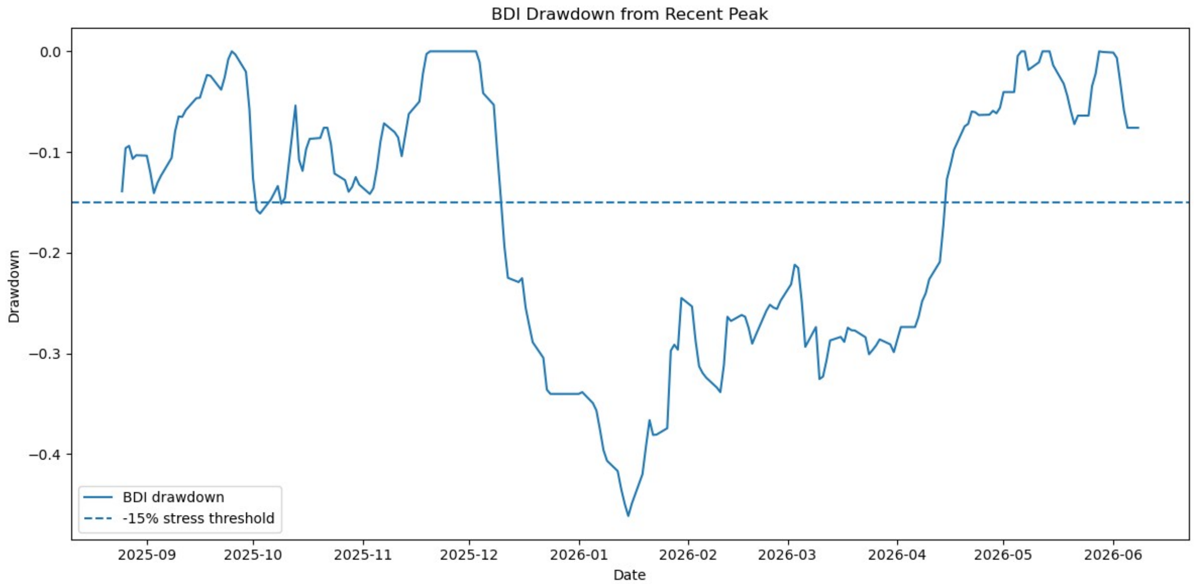



Volatility has moved a lot during the sample. It was much higher during the earlier stress period, especially around the sharp decline in early 2026.

Current 20-day annualized volatility is around 31.2%. That is not very low, but it is far below the highest levels in the sample. The current market therefore does not look like a volatility shock.

### 6.3 Drawdown

*Figure 3: BDI drawdown from recent peak.*

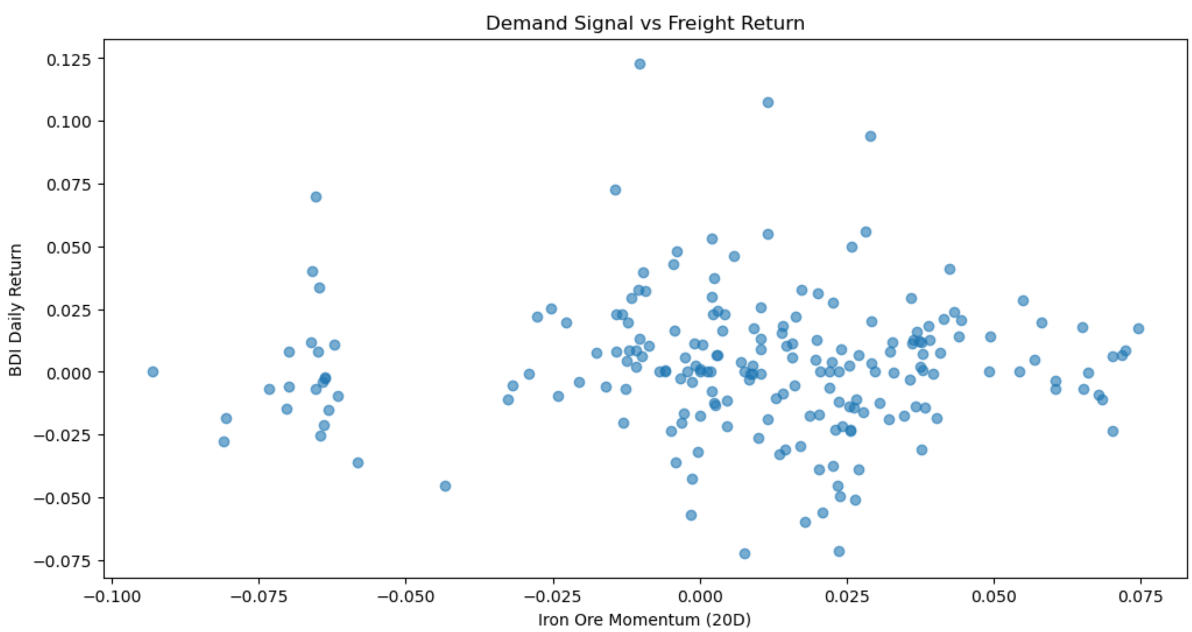



The drawdown chart turns the BDI movement into a simple risk view. Earlier in the sample, BDI fell more than 40% from its previous peak. Compared with that, the current drawdown of around -7.6% is moderate.

This supports the same interpretation as the volatility chart: the market has weakened, but it has not entered a severe stress phase.

### 6.4 Iron ore demand proxy

*Figure 4: Iron ore momentum versus BDI daily returns.*




This is the most important chart in the note. If iron ore momentum were a strong short-term driver of BDI returns, we would expect to see a clearer upward pattern. The plot does not show that.

The correlation confirms this. The correlation between BDI daily returns and 20-day iron ore momentum is approximately -0.002, which is effectively zero in this sample.

*Table 2: Correlation table.*

| | BDI_ret | IronOre_momentum20 |
|---|---|---|
| BDI_ret | 1 | -0.002 |
| IronOre_momentum20 | -0.002 | 1 |

This does not mean iron ore is unimportant. It means that iron ore momentum alone is not enough to explain short-term freight rate movements.

## 7. Key Findings

Finding 1: Iron ore matters economically, but is not enough statistically.

Iron ore is clearly relevant to dry bulk shipping. But in this recent sample, the relationship between 20-day iron ore momentum and daily BDI returns is weak. Iron ore should therefore be part of the market discussion, but it should not be treated as a standalone explanation.

Finding 2: Dry bulk is a multi-driver market.

The weak short-term relationship suggests that freight markets are driven by several forces at the same time. Missing drivers may include vessel supply, port congestion, fleet utilization, seasonality, route-specific cargo flows, regional demand differences and fuel cost dynamics.

Finding 3: The current market is neutral.

The medium-term trend is still supportive, but short-term momentum has weakened. The market is not in stress, but it should be monitored carefully.

## 8. Commercial Interpretation

For a trading or commercial analytics team, the value of this framework is not that it gives a direct trading recommendation. It does not say buy, sell or hold. Its value is that it gives a clear structure for discussing the market.

The framework separates the market view into four parts:

- Freight trend
- Volatility and drawdown risk
- Demand-side commodity signal
- Cost-side energy signal

This makes the analysis more disciplined. Instead of only saying that BDI is up or iron ore is down, the framework asks: Is the freight trend still positive? Has short-term momentum weakened? Is the market close to a stress state? Is there still demand-side support? What is happening to energy cost pressure?

A trader or commercial stakeholder would not use this as a standalone decision rule. But it can help frame the discussion before making a decision. It also shows where the current evidence is not enough. In this case, the next useful step is to add supply-side and vessel-class information before drawing stronger conclusions.

The current interpretation is: the dry bulk market is still supported in the medium term, but weakening iron ore momentum and a short-term BDI pullback mean the market should be watched more carefully. It is not yet a stress regime, but the balance of signals is less supportive than before.

## 9. Limitations

This framework is intentionally simple. It should not be read as a validated forecasting model.

The main limitations are:

- BDI is an index, not a direct trading position. It gives a broad market signal, but it does not represent a specific vessel, route, contract or exposure.
- Iron ore is only one demand proxy. Dry bulk shipping also depends on coal, grain, minor bulks and regional cargo flows.
- Supply-side variables are missing. The current version does not include vessel availability, fleet utilization, port congestion or route-specific capacity.
- The sample is short. The analysis mainly uses around one recent year of data. It can support a short-term view of the recent market period, but it cannot show whether the relationship is stable across long cycles or historical stress regimes.
- Major historical stress periods are not tested. The sample does not cover the initial COVID-19 supply-chain shock or the first phase of the Russia-Ukraine energy shock, so the conclusion should not be generalized to all freight-market regimes.
- Forward-filling may smooth some short-term moves, because the markets have different holiday calendars.
- The framework is diagnostic, not predictive. It describes the current market state but does not claim to forecast future freight rates.

These limitations are not just weaknesses. They are part of the output. They show what the framework can say, and what it cannot say yet.

## 10. Next Research Agenda

The main result of Version 1 is that one demand-side proxy is not enough. The natural next step is to extend the framework from a demand-and-cost view to a broader demand-supply-risk view.

Version 2 should consider:

- Vessel-class indices such as Capesize and Panamax
- Route-level freight indicators
- Port congestion indicators
- Seasonality controls
- Additional commodity demand proxies
- A longer historical sample
- Weekly-frequency tests, to reduce noise in daily data

The next research question would be: Does adding supply-side and vessel-class information improve the explanation of freight market regimes?

From a data science perspective, the next practical step would also be to make the workflow more reproducible: cleaner Python scripts, a documented data pipeline, and later possibly a simple orchestration or containerized workflow. A small machine-learning extension could also be tested, for example using the indicators to classify market regimes or compare simple forecasting models. That should only be done after the data window is extended, because the current sample is too short for a reliable machine-learning conclusion.

## 11. Conclusion

This note shows how a small set of public market data can be used to build a simple and transparent dry bulk market diagnostic framework.

The main finding is clear: iron ore is economically important, but iron ore momentum alone does not explain short-term BDI movements in this recent sample. Dry bulk freight markets need a broader view that includes demand, supply, cost and market structure.

The current market is best described as Neutral: the medium-term trend is still supportive, but the short-term picture has weakened.

The main value of the project is the process: start with a clear market hypothesis, test it with data, interpret the result carefully, state the limitations, and define the next useful step.